

SQL Hands-on Practice: DDL, DML, DCL, TCL

Database: Bank Management System

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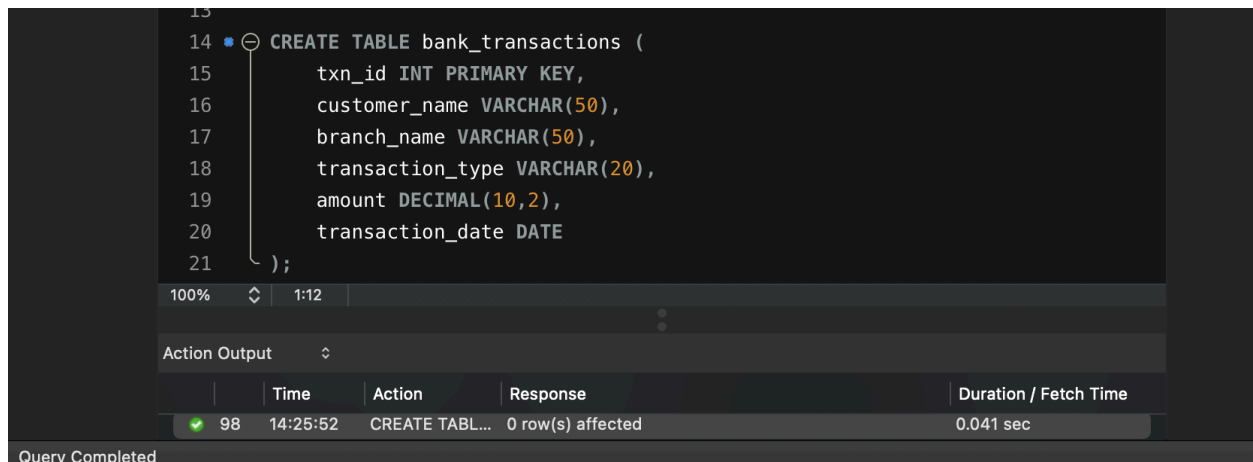
Roll No: 2520030237

Section: 7

SECTION 1 — DDL (Data Definition Language)

Step 1 — CREATE TABLE

```
CREATE TABLE bank_transactions (  
    txn_id INT PRIMARY KEY,  
    customer_name VARCHAR(50),  
    branch_name VARCHAR(50),  
    transaction_type VARCHAR(20),  
    amount DECIMAL(10,2),  
    transaction_date DATE  
);
```



Step 2 — ALTER TABLE — ADD COLUMN

```
ALTER TABLE bank_transactions  
ADD account_no VARCHAR(20);
```

```
18
19 ALTER TABLE bank_transactions
20 ADD account_no VARCHAR(20);
21
```

00% 1:16

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 99	14:27:14	ALTER TABLE...	0 row(s) affected Records: 0 Duplicates: 0 Warnings...	0.036 sec

Step 3 — ALTER TABLE — MODIFY COLUMN

```
ALTER TABLE bank_transactions
MODIFY customer_name VARCHAR(100);
```

```
17
18
19 ALTER TABLE bank_transactions
20 MODIFY customer_name VARCHAR(100);
21
```

100% 1:18

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 100	14:27:48	ALTER TABLE...	0 row(s) affected Records: 0 Duplicates: 0 Warnings...	0.018 sec

Step 4 — TRUNCATE TABLE

```
TRUNCATE TABLE bank_transactions;
```

```
17
18
19
20 TRUNCATE TABLE bank_transactions;
```

100% 1:16

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 101	14:28:29	TRUNCATE T...	0 row(s) affected	0.0091 sec

Completed

Step 5 — DROP TABLE

```
DROP TABLE bank_backup;
```

17
18 CREATE TABLE bank_backup (id INT);
19 DROP TABLE bank_backup;

100% 1:17

Action Output

	Time	Action	Response	Duration / Fetch Time
102	14:29:22	CREATE TABL...	0 row(s) affected	0.0083 sec
103	14:29:37	DROP TABLE...	0 row(s) affected	0.012 sec

Completed

SECTION 2 — DML (Data Manipulation Language)

Step 6 — Insert Sample Data (10 rows)

```
INSERT INTO bank_transactions VALUES
(101, 'Ravi', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
(102, 'Sita', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
(103, 'Kiran', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
(104, 'Anil', 'Vizag', 'Deposit', 8000, '2024-01-10'),
(105, 'Priya', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
(106, 'Ramesh', 'Vizag', 'Deposit', 15000, '2024-01-12'),
(107, 'Keerthi', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
(108, 'Rahul', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
(109, 'Sneha', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
(110, 'Madhu', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
```

100%

33:50

1 error found

Result Grid

Filter Rows:

Search

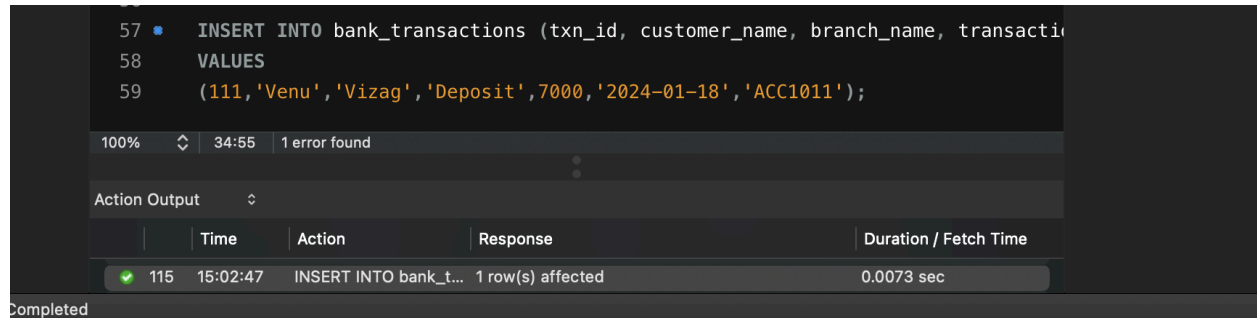
Edit:

Export/Import:

txn_id	customer_name	branch_name	transaction_ty...	amount	transaction_d...	account_no
101	HANSIKA	Hyderabad	Deposit	5000.00	2024-01-05	ACC1001
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	ACC1002
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	ACC1003
104	Anil	Vizag	Deposit	8000.00	2024-01-10	ACC1004
105	Priya	Hyderabad	Withdrawal	3500.00	2024-01-11	ACC1005
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	ACC1006
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	ACC1007
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	ACC1008
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	ACC1009
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	ACC1010
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Step 7 — INSERT

```
INSERT INTO bank_transactions
VALUES
(111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18');
```



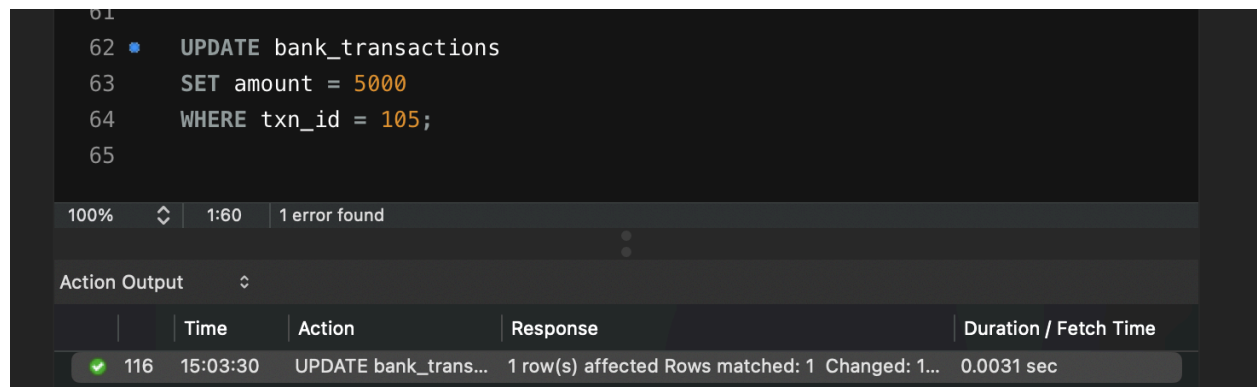
The screenshot shows a SQL IDE interface. The top pane displays the SQL code for Step 7: `INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date) VALUES (111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18', 'ACC1011');`. The bottom pane shows the execution results. A progress bar indicates 100% completion. Below it, the 'Action Output' section shows a table with columns: Time, Action, Response, and Duration / Fetch Time. The table contains one row: 115, 15:02:47, INSERT INTO bank_t..., 1 row(s) affected, 0.0073 sec. The status 'Completed' is shown at the bottom left.

	Time	Action	Response	Duration / Fetch Time
✓ 115	15:02:47	INSERT INTO bank_t...	1 row(s) affected	0.0073 sec

Completed

Step 8 — UPDATE

```
UPDATE bank_transactions
SET amount = 5000
WHERE txn_id = 105;
```



The screenshot shows a SQL IDE interface. The top pane displays the SQL code for Step 8: `UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105;`. The bottom pane shows the execution results. A progress bar indicates 100% completion. Below it, the 'Action Output' section shows a table with columns: Time, Action, Response, and Duration / Fetch Time. The table contains one row: 116, 15:03:30, UPDATE bank_trans..., 1 row(s) affected Rows matched: 1 Changed: 1..., 0.0031 sec. The status 'Completed' is shown at the bottom left.

	Time	Action	Response	Duration / Fetch Time
✓ 116	15:03:30	UPDATE bank_trans...	1 row(s) affected Rows matched: 1 Changed: 1...	0.0031 sec

Completed

Step 9 — DELETE

```
DELETE FROM bank_transactions
WHERE txn_id = 111;
```

```

66
67 • DELETE FROM bank_transactions
68 WHERE txn_id = 111;
69
100% 20:64 1 error found

```

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 117	15:03:56	DELETE FROM bank...	1 row(s) affected	0.0019 sec

Step 10 — SELECT (All Rows)

```

SELECT *
FROM bank_transactions;

```

```

70
71
72 • SELECT *
73 FROM bank_transactions;
74
100% 20:68 1 error found

```

Result Grid

	txn_id	customer_name	branch_name	transaction_ty...	amount	transaction_d...	account_no
	101	HANSIKA	Hyderabad	Deposit	5000.00	2024-01-05	ACC1001
	102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	ACC1002
	103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	ACC1003
	104	Anil	Vizag	Deposit	8000.00	2024-01-10	ACC1004
	105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	ACC1005
	106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	ACC1006
	107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	ACC1007
	108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	ACC1008
	109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	ACC1009
	110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	ACC1010
	NULL	NULL	NULL	NULL	NULL	NULL	NULL

bank_transactions 4

Apply Revert

Step 11 — SELECT WITH WHERE

```

SELECT *
FROM bank_transactions
WHERE transaction_type='Deposit';

```

72
73 • SELECT *
74 FROM bank_transactions
75 WHERE transaction_type='Deposit';
76
77

100% 24:71 1 error found

Result Grid Filter Rows: Search Edit: Export/Import:

txn_id	customer_name	branch_name	transaction_ty...	amount	transaction_d...	account_no
101	HANSIKA	Hyderabad	Deposit	5000.00	2024-01-05	ACC1001
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	ACC1003
104	Anil	Vizag	Deposit	8000.00	2024-01-10	ACC1004
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	ACC1006
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	ACC1008
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	ACC1010
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Result Grid
Form Editor
Field Types

Step 12 — SELECT WITH ORDER BY

```
SELECT *
FROM bank_transactions
ORDER BY amount DESC;
```

76
77 • SELECT *
78 FROM bank_transactions
79 ORDER BY amount DESC;
80
81

100% 23:74 1 error found

Result Grid Filter Rows: Search Edit: Export/Import:

txn_id	customer_name	branch_name	transaction_ty...	amount	transaction_d...	account_no
106	Hamesh	Vizag	Deposit	15000.00	2024-01-12	ACC1006
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	ACC1003
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	ACC1010
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	ACC1008
104	Anil	Vizag	Deposit	8000.00	2024-01-10	ACC1004
101	HANSIKA	Hyderabad	Deposit	5000.00	2024-01-05	ACC1001
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	ACC1005
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	ACC1009
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	ACC1002
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	ACC1007
NULL	NULL	NULL	NULL	NULL	NULL	NULL

bank_transactions 6 Apply Revert

Result Grid
Form Editor
Field Types

SECTION 3 — DCL (Data Control Language)

Step 13 — GRANT SELECT

```
GRANT SELECT
ON bank_transactions
TO Auditor1;
```

The screenshot shows the SQL Server Enterprise Manager interface. The SQL text area contains the following code:

```
88
89 GRANT SELECT
90 ON bank_transactions
91 TO 'Auditor1'@'localhost';
92
```

Below the SQL text, the status bar indicates "100%", "59:82", and "1 error found". The "Action Output" pane is expanded, showing a table with the following data:

	Time	Action	Response	Duration / Fetch Time
✓ 125	15:08:50	GRANT SELECT ON...	0 row(s) affected	0.0027 sec

The status bar at the bottom indicates "Completed".

Step 14 — GRANT ALL PRIVILEGES

```
GRANT ALL
ON bank_transactions
TO BranchManager;
```

The screenshot shows the SQL Server Enterprise Manager interface. The SQL text area contains the following code:

```
92
93
94 CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY 'Pass@123';
95 GRANT ALL
96 ON bank_transactions
97 TO 'BranchManager'@'localhost';
98
99
```

Below the SQL text, the status bar indicates "100%", "1:93", and "1 error found". The "Action Output" pane is expanded, showing a table with the following data:

	Time	Action	Response	Duration / Fetch Time
✓ 128	15:10:29	GRANT ALL ON ban...	0 row(s) affected	0.0031 sec

The status bar at the bottom indicates "Completed".

Step 15 — REVOKE SELECT

```
REVOKE SELECT
ON bank_transactions
FROM Auditor1;
```

```
98
99 REVOKE SELECT
100 ON bank_transactions
101 FROM 'Auditor1'@'localhost';
102
```

100% 1:98 1 error found

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 129	15:11:17	REVOKE SELECT ON...	0 row(s) affected	0.0021 sec

deleted

Step 16 — REVOKE ALL

```
REVOKE ALL
ON bank_transactions
FROM BranchManager;
```

```
REVOKE ALL PRIVILEGES
ON bank_transactions
FROM 'BranchManager'@'localhost';
```

1:102 2 errors found

Output

	Time	Action	Response	Duration / Fetch Time
30	15:13:54	REVOKE ALL PRIVIL...	0 row(s) affected	0.0029 sec

SECTION 4 — TCL (Transaction Control Language)

Step 17 — COMMIT

```
UPDATE bank_transactions
SET amount=6000
WHERE txn_id=101;

COMMIT;
```



```
110
111 • UPDATE bank_transactions
112   SET amount=6000
113   WHERE txn_id=101;
114
115 • COMMIT;
116
```

100% 1:116 2 errors found

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 128	15:10:29	GRANT ALL ON ban...	0 row(s) affected	0.0031 sec

Completed

Step 18 — SAVEPOINT

```
SAVEPOINT Before_Update;
```

```
115 • COMMIT;
116
117 • SAVEPOINT Before_Update;
118
```

100% 18:113 2 errors found

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 133	15:15:10	SAVEPOINT Before_...	0 row(s) affected	0.00030 sec

Completed

Step 19 — ROLLBACK TO SAVEPOINT

```
UPDATE bank_transactions
SET amount=99999
WHERE txn_id=102;

ROLLBACK TO Before_Update;

SELECT * FROM bank_transactions WHERE txn_id=102;
```

Server Status

Client Connections

Users and Roles

Status and Performance

Data Export

Data Import

Object Explorer

No object selected

128

129

130

131

132

133

134

135

136

137

138

139

START TRANSACTION;

SAVEPOINT Before_Update;

UPDATE bank_transactions

SET amount=99999

WHERE txn_id=102;

ROLLBACK TO Before_Update;

SELECT * FROM bank_transactions WHERE txn_id=102;

COMMIT;

100%

1:128

2 errors found

Result Grid

Filter Rows: Search

Edit: [Icons]

Export/Import: [Icons]

Result Grid

Form Editor

txn_id	customer_name	branch_name	transaction_ty...	amount	transaction_d...	account_no
102	Sita	Hyderabad	Withdrawal	99999.00	2024-01-06	ACC1002

Step 20 — Full TCL Example

```

UPDATE bank_transactions
SET amount=7000
WHERE txn_id=101;

SAVEPOINT SP1;

UPDATE bank_transactions
SET amount=9000
WHERE txn_id=102;

ROLLBACK TO SP1;

COMMIT;

SELECT * FROM bank_transactions WHERE txn_id IN (101,102);

```

Client Connections

Users and Roles

Status and Performance

Data Export

Data Import

Object Explorer

No object selected

143

144

145

146

147

148

149

150

151

152

153

START TRANSACTION;

UPDATE bank_transactions

SET amount=7000

WHERE txn_id=101;

SAVEPOINT SP1;

UPDATE bank_transactions

SET amount=9000

WHERE txn_id=102;

ROLLBACK TO SP1;

COMMIT;

SELECT * FROM bank_transactions WHERE txn_id IN (101,102);

100%

19:143

2 errors found

Result Grid

Filter Rows: Search

Edit: [Icons]

Export/Import: [Icons]

Result Grid

Form Editor

txn_id	customer_name	branch_name	transaction_ty...	amount	transaction_d...	account_no
101	HANSIKA	Hyderabad	Deposit	7000.00	2024-01-05	ACC1001
102	Sita	Hyderabad	Withdrawal	9000.00	2024-01-06	ACC1002

Automatic color disabled. Use manually get current caret position to toggle auto